



PERTH MODERN SCHOOL
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Independent Public School

Course Methods

Year 11

Student name: _____

Teacher name: _____

Date: 17/02/20

Task type: Response

Time allowed for this task: 40 mins

Number of questions: 6

Materials required: NO CALCULATOR REQUIRED
NO NOTES REQUIRED

Standard items: Pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: Drawing instruments, templates and formula sheet

Marks available: 37 marks

Task weighting: 10 %

Formula sheet provided: Yes

Note: All part questions worth more than 2 marks require working to obtain full marks.

Question 1 (1.1.6)**(2, 2 = 4 marks)**Solve each of the following for x .

i) $2x - 3 = 11 - 5x$

ii) $10 - 2x = \frac{2x}{3}$

Question 2 (1.1.4, 11.5, 1.1.6)**(2, 3, 2, 3 = 10 marks)**

Determine the equation of a line that passes through the point $(-4, 6)$ and:

- i) has a gradient of 3

- ii) passes through the point $(2, 5)$.

- iii) is parallel to the line $2y - 4x = -7$.

- iv) is perpendicular to the line $2y - x - 8 = 0$.

Question 3 (1.1.1, 1.1.5, 1.1.6)**(3, 2, 2 = 7 marks)**

The coordinates $P(2, p)$ and $Q(q + 1, 3q - 2)$ both lie on the line $y = 5x + 1$.

a) Find:

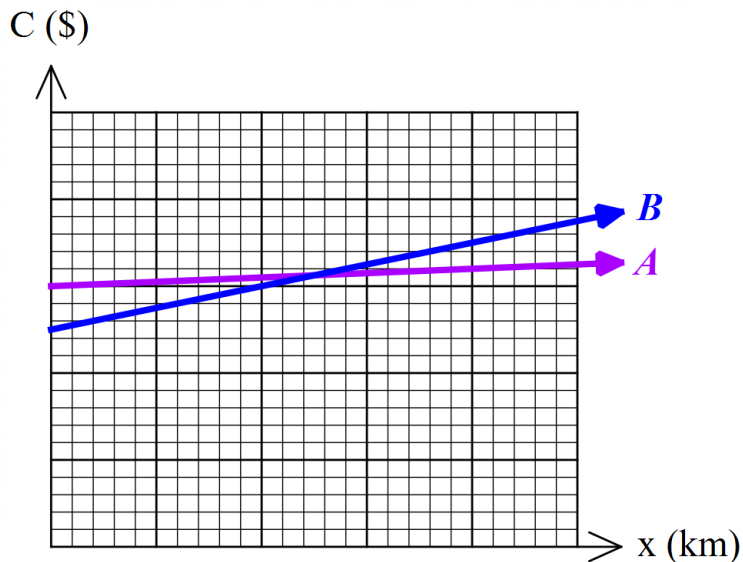
i) the values of p and q .

ii) the midpoint of PQ .

b) For what value of m does the line $y = mx + 2$ not intersect with the line $y = 5x + 1$?
Justify your answer.

Question 4 (1.1.4, 1.1.5)**(2, 1, 1, 2, 1 = 7 marks)**

The graph below shows cost, C , in dollars versus distance x , in kilometres, for two different car rental companies A and B. (Assume that parts of distance are charged for proportionately.)



The costs for each company are outlined in the table below.

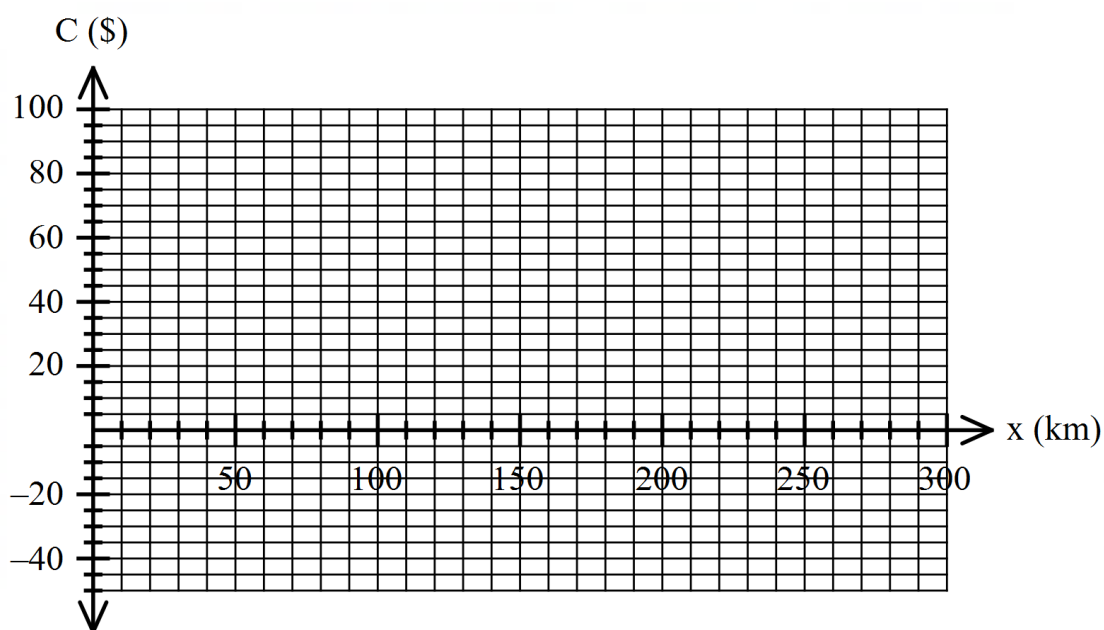
- a) Which cost equation corresponds to Company A and Company B?

$C = 250 + 0.25x$	$C = 300 + 0.05x$

- b) Explain what the gradient in the equation $C = 250 + 0.25x$ represents.

- c) Construct a linear rule for $y = C_A - C_B$, the difference in cost between Company A and Company B.

- d) Sketch the equation from part c) on the graph below clearly showing all intercepts.



- e) Using the graph in part d) determine the number of km when the costs of Company A is cheaper than those of Company B .

Question 5 (1.1.6)**(5 marks)**

Solve for x , expressing your answer in its simplest form in terms of a and/or b .

$$\frac{x+a}{b} = \frac{b-x}{a} \quad [a, b \neq 0]$$

(2, 1, 1 = 4 marks)

a) Construct a linear equation for t using the information given.

b) Solve your equation in part a) and hence calculate the value of t .

c) Find the distance between A and B .

END OF TEST